Faithfulness Under Pressure

Context and History Pruning for Conversational RAG Faithfulness

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Problem

Question

When a small language model receives long, noisy retrieved context, can pruning make its answer more faithful to the evidence?

- Task: conversational open-domain question answering with RAG.
- Main benchmark: QReCC, where each question depends on previous conversation turns.
- Output: short answer generated from retrieved/pruned evidence and retained dialogue history.
- Main risk: the model may keep stale history, miss the current evidence, or hallucinate unsupported claims.

Why Pruning Matters

Long context is not free.

- More tokens increase latency and memory cost.
- Retrieved passages and old dialogue turns often contain near-misses, stale entities, and distractors.
- Small LMs are especially sensitive to noisy evidence.

Source: Liu et al., TACL 2024, "Lost in the Middle".[1]

Long context is not always used well.

- Liu et al. show that models often perform best when relevant information is near the beginning or end of the context.
- Performance can drop when evidence is in the middle, even for long-context models.

[1] https://aclanthology.org/2024.tacl-1.9/

Project Pipeline

QReCC conversational turn + history

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Optional rewrite / current-turn query

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Retriever: top-k passages

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Context + history pruner

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Mistral-7B-Instruct Generator

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LLM-as-Judge Faithfulness Scoring

- The retriever and generator are held fixed across pruning strategies.
- Only retrieved context and retained history change.
- This isolates the effect of pruning on faithfulness, answer quality, and compression.

Pruning Strategies

ID	Strategy	Role
B1	No pruning	Full retrieved context; maximum evidence and maximum noise.
B2	Naive truncation	Hard token budget; common but relevance-agnostic.
B3	RECOMP-style extractive	Selects query-relevant sentences from retrieved passages.
S1	LLMLingua-2	Token-level compression using a bidirectional encoder classifier.
S2	Provence	Sentence-level sequence-labeling pruner, also related to reranking.
S3	History pruning	Removes redundant context using conversation history.
S4	Combined	Sentence pruning plus history-aware pruning.
S5	CfC/LNN extension	Exploratory adaptive pruning model for multi-turn context.

Related Work Signals

- **RECOMP**: compresses retrieved documents before generation; reports compression rates as low as 6% with minimal performance loss in evaluated settings.[2]
- **LLMLingua-2**: formulates prompt compression as token classification and reports 1.6–2.9$\times$ end-to-end latency acceleration at 2–5$\times$ compression.[3]
- **Provence**: treats context pruning as sequence labeling and reports negligible to no performance drop at almost no additional RAG pipeline cost.[4]

Our angle

We use QReCC because the central problem is conversational: deciding which past turns and retrieved passages still matter for the current question.

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[2] https://proceedings.iclr.cc/paper_files/paper/2024/hash/bda88ed2892f5e61c9a9bf215c566913-Abstract-Conference.html

[3] https://huggingface.co/papers/2403.12968

[4] https://arxiv.org/abs/2501.16214

Dataset and Experimental Setup

Main benchmark: QReCC

- 14K conversations.
- About 81K question-answer pairs.
- Built for question rewriting in conversational context.
- Retrieval collection: 10M web pages split into 54M passages.
- Each sample includes original question, context, rewrite, answer, and answer provenance.

Current experiment status

- Faithfulness: fraction of generated claims supported by context.
- EM/F1: lexical answer correctness against gold answer.
- Compression ratio: output tokens / input tokens.
- History retention: retained turns / original turns.
- Pilot HotpotQA runs remain only as implementation sanity checks.

Source: Anantha et al., NAACL 2021 / QReCC dataset release.[5]

[5] https://aclanthology.org/2021.naacl-main.44/

Pilot Results: HotpotQA Sanity Run

Strategy	Faith.	EM	F1	Comp.	Lat. s
B1: No pruning	0.718	0.040	0.148	1.000	6.76
B2: Naive truncation	0.621	0.040	0.139	0.918	4.95
B3: RECOMP	**0.750**	0.040	0.141	0.911	5.38
S1: LLMLingua-2 fallback	**0.750**	0.040	0.141	0.911	5.38
S2: Provence	0.657	0.000	0.057	**0.072**	**4.49**
S3: History, empty	0.718	0.040	0.148	1.000	6.34
S4: Combined, empty	0.657	0.000	0.057	**0.072**	**4.48**

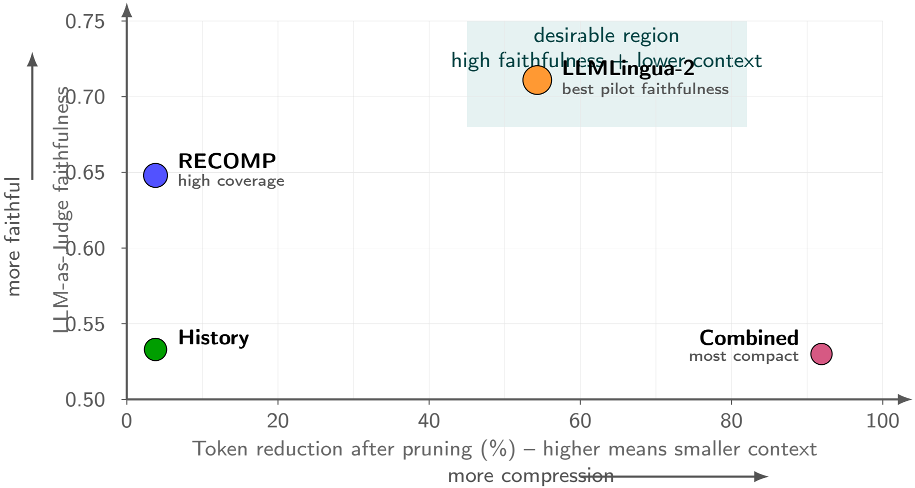
Notebook 2, 50 HotpotQA validation samples. These are pilot implementation checks, not the final QReCC benchmark results.

Pilot Results: Gap-Closing Run

Strategy	Faith.	EM	F1	Comp.	Cov.	Noise
B3: RECOMP	0.648	0.080	0.195	0.962	**0.760**	0.624
S1: LLMLingua-2 real	**0.711**	0.040	0.167	0.457	0.000	0.792
S3: History two-turn	0.533	**0.100**	**0.214**	0.962	0.627	0.641
S4: Combined two-turn	0.530	0.040	0.112	**0.081**	0.222	**0.284**

- LLMLingua-2 is the strongest faithfulness result in this pilot run.
- Combined pruning is extremely compact and low-noise, but loses too much support.
- QReCC is the correct next benchmark for testing real history pruning.

Faithfulness–Compression Trade-off



Current numbers are HotpotQA pilot outputs; final benchmark target is QReCC conversational QA.

What We Learned

- **Pruning can improve faithfulness**, but not every pruning method helps equally.
- **Aggressive compression is dangerous for conversational QA**: if the relevant history turn disappears, the current question becomes ambiguous.
- **Faithfulness and correctness are different**: a faithful answer can still be the wrong answer, and a correct short answer can be judged poorly.
- **Exact coverage is too strict for token compression**: LLMLingua-2 may fragment evidence so gold sentences no longer match verbatim.
- **Conversation history is not just extra text**: old turns can help or distract depending on the current question.

Error Analysis Examples

Error type	Example pattern
Evidence removed	The "Arizona" question loses the sentence saying the song was written by Kenny Young.
Verbose correct answer	For yes/no questions, the model may answer correctly with explanation but get EM $= 0$.
Judge false negative	A short correct phrase such as "Political correctness" can be scored as unfaithful.
Entity near-miss	The generator outputs "T. M. Howard" instead of "T. R. M. Howard".

Takeaway

We need judge calibration and qualitative analysis before treating one metric as ground truth.

CfC Extension: The Core Idea

Where the idea came from

When reading dense, information-rich text, humans slow down. High-entropy words take longer to process because they carry more unexpected information. Can we use this same signal to decide which conversation turns deserve more "processing time" in a neural network?

- Fixed thresholds do not know whether a turn is easy, noisy, redundant, or depends on earlier dialogue.
- We explore Closed-form Continuous-time networks (CfCs): temporal dynamics without a heavy ODE solver.[6]
- Key insight: map per-turn *surprisal* to a continuous time step Δt_i, so surprising turns get a wider integration window in the network's ODE.

[6] https://www.nature.com/articles/s42256-022-00556-7

Cognitive Motivation: Reading Time $\propto$ Surprisal

Surprisal theory (psycholinguistics)

Reading time on a word is proportional to its surprisal $-\log P(w_j \mid w_{<j})$. A word that violates expectations takes more cognitive effort to integrate.

- Levy (2008): syntactic surprisal predicts word-by-word reading times.
- Smith & Levy (2013): the relationship is *logarithmic* and holds across many corpora.
- **Transfer to NLP:** if a turn is lexically surprising (high surprisal from a proxy LM), it likely introduces new information — and is more likely to be important for answering the current question.
- We do not claim surprisal *equals* importance; we ask whether it is a *useful proxy* — and test this with a simulation before training.

Levy 2008, Cognition 106(3). Smith & Levy 2013, Cognition 128(3).

From Reading Time to the CfC Time Step

For utterance u_i, compute surprisal with DistilGPT-2 (proxy LM):

$$S(u_i) = -\frac{1}{N}\sum_{j=1}^{N} \log P(w_j \mid w_{<j})$$

Map surprisal to a continuous time step (rank-normalised for outlier robustness):

$$\Delta t_i = \Delta t_{\text{min}} + \beta \cdot \frac{\text{rank}(S_i)}{T}$$

The CfC network then integrates over Δt_i at each turn. The *commitment fraction* measures how strongly the state “locks on”:

$$\varphi_i = 1 - e^{-\lambda \Delta t_i}$$

- Large $\Delta t_i \to$ large $\varphi_i \to$ turn leaves a stronger trace in the hidden state $\to$ higher importance score.
- Rank normalisation prevents one outlier surprisal from dominating the time budget (validated by Simulation S5).

Why Simulations Before Training?

The risk

- Surprisal may not correlate with true turn importance on QReCC at all.
- If the correlation $\rho \approx 0$, entropy adds no signal and training is wasted.
- Hyperparameters β, Δt_{min}, τ affect faithfulness but are hard to tune end-to-end.

The strategy

- Seven model-free simulations, each answering one design question analytically.
- Each simulation outputs a concrete config value or a go/no-go threshold.
- Only if all sims pass do we commit to teacher-label generation and CfC training.

Go/no-go condition (S3)

Measure ρ(surprisal, teacher-importance) on real QReCC *before* training. If $\rho < 0.1$: entropy adds nothing over uniform. If $\rho \geq 0.5$: entropy beats cosine similarity.

Simulation Suite: 7 Design Questions Answered

#	Question	Result
S0	Does analytic LTC commitment transfer to real ncps CfC?	$\rho = +0.944$; error $\approx 4 \times 10^{-10}$
S1	What is the analytic commitment–surprisal curve?	$d\varphi/dS = \lambda\beta(1-\varphi)$; 31.6% turns saturate
S2	Which β^* and Δt^*_{min} maximise importance recovery?	$\beta^* = 1.0$, $\Delta t^*_{\text{min}} = 0.05$
S3	What ρ threshold makes entropy useful? (go/no-go)	$\rho^* \approx 0.1$ vs. uniform; $\rho \geq 0.5$ beats cosine
S4	Does LOO-KL measure raw importance or marginal necessity?	KL = marginal necessity; correct pruning target
S5	Which Δt mapping is robust to outlier surprisal?	Rank normalisation most robust
S6	What compression–fidelity trade-off does the system achieve?	40% compression at 90% fidelity; uniform 9%, cosine 31%

Entropy-Driven CfC Pruning: Architecture

Simulation outputs $\rightarrow$ model design.

- **Input per turn:** SBERT embedding (dim 384) + rank-normalised Δt_i from DistilGPT-2 surprisal.
- **Hidden dynamics:** CfC cell; state h_i integrates over Δt_i (wider window = stronger commitment).
- **Per-turn readout:** $\hat{p}_i = \sigma(Wh_i + b)$, importance $\in [0, 1]$.
- **Teacher labels:** leave-one-out KL divergence on Mistral-7B-Instruct output distributions.
- **Loss:** MSE between $\hat{p}_i$ and normalised teacher KL.
- **Pruning:** discard turns where $\hat{p}_i < \tau^* \approx 0.48$ (from S6: 90% fidelity target).

$\Delta t_{\text{min}} = 0.05 \quad \beta = 1.0 \quad \tau^* \approx 0.48$
(all three derived from simulations, not hand-tuned)

CfC Evaluation Plan

Baselines

- Full context.
- Random pruning.
- Cosine-similarity history pruning.
- Existing sentence/token LNN pruners.

Metrics

- ROUGE-L and BERTScore.
- Faithfulness with LLM-as-Judge.
- Token reduction.
- Time to first token.
- History/evidence coverage.

Current status

The CfC/LNN code and Colab notebooks are implemented as an exploratory extension, but final quantitative CfC results are not complete yet.

Contributions So Far

- Built a modular RAG pruning harness with shared pruner interface.
- Implemented no pruning, truncation, RECOMP-style extraction, LLMLingua-2, Provence, history pruning, and combined pruning.
- Ran baseline and corrected gap-closing Colab experiments.
- Added coverage/noise metrics to diagnose pruning quality.
- Exported qualitative error-analysis candidates.
- Designed the entropy-driven CfC pruner: cognitive motivation, math derivation, and 7-simulation verification suite before training.
- Derived all hyperparameters (β^*, Δt^*_{min}, τ^*, mapping) analytically from simulations.

Remaining Work

- Complete human annotation and report Cohen's κ for judge calibration.
- Port the final evaluation split from HotpotQA pilot runs to QReCC.
- Tune Provence and combined-pruning thresholds for conversational evidence coverage.
- Replace exact supporting-sentence coverage with a semantic history/evidence coverage metric.
- Finish CfC teacher-label generation and training.
- Run CfC against full context, random pruning, and cosine-history baselines.
- Evaluate on real QReCC dialogue histories instead of synthetic two-turn histories.

Conclusion

Main finding

Context pruning is useful, but the best pruner is not simply the most aggressive one.

- LLMLingua-2 gave the best corrected faithfulness score in our pilot runs.
- Combined pruning showed the strongest compression/noise reduction in pilot runs, but removed too much supporting evidence.
- CfC-style adaptive pruning is our next step for deciding *when* context should be preserved, compressed, or forgotten.

References

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